Laboratory work No. 2. Decision Trees and Random Forest.

1) **Dataset**
   a) Use the data set analyzed in the first laboratory work (Table 1 contains references, Table 2 contains the internal code, dataset title, predicted feature).
   b) Split data set into the test data set (at least 10%, will only be used in the last step of model building) and the training and validation data sets. Prepare training and validation sets for training (remove outliers, handle missing values, remove uninformative features, etc.).

2) **Regression Decision Tree**
   a) Implement the algorithm for creating a decision tree for regression. The decision tree must handle categorical features directly (without converting them to numerical values); categorical features are evaluated by creating a separate node for each category (the resulting tree is not binary). For numerical variables, at least five threshold values are tested. For binary variables (true/false; 1/0), two branches are created immediately.
   b) With the training dataset, use at least five combinations of the values of the decision tree parameters (tree depth, number of splits, etc.) with the implemented algorithm, calculate and in a **table** provide the model error metrics (MSE, MAE, MAPE, etc.) for training and validation datasets. What tree parameters should be used? Justify your answer.
   c) Implement a regression decision tree model using the scikit-learn library function DecisionTreeRegressor, in a **table** provide the error metrics with the parameter sets used in part b) for the test and validation datasets. Compare the results of parts b and c.
   d) Plot the tree (criteria applied) that has the smallest errors for the validation dataset or print the features (criteria) used to make decisions in the first three layers.

3) **Classification Decision Tree**
   a) Map the continuous target variable to categorical by creating 5 classes (e.g., price is very small (0), small (1), medium (2), large (3), very large (4)).
   b) Implement the classification decision tree construction algorithm. The decision tree must handle categorical features directly (without converting them to numerical values); categorical features are evaluated by creating a separate node for each category (the resulting tree is not binary). For numerical variables, at least five threshold values are tested. For binary variables (true/false; 1/0), two branches are created immediately.
   c) With the training dataset, use at least five combinations of values for the decision tree parameters (tree depth, number of splits, etc.), calculate and in a **table** provide the model error metrics (accuracy, precision, sensitivity, specificity, etc.) for the training and validation datasets. What tree parameters should be used? Justify your answer. For the selected model, provide the confusion matrices for the training and validation datasets.
   d) Implement a classification decision tree model using the DecisionTreeClassifier function of the scikit-learn library, provide error metrics with the parameter sets used in part c) for the test and validation datasets. Compare the results of parts c and d.
   e) Plot the chosen classification tree (criteria applied) or print the features (criteria) used to make decisions in the first three layers.

4) **Regression Random Forest**
   a) Implement the random forest algorithm (use the regression decision tree implementation).
   b) Choose 3 different numbers of trees (e.g. 5, 10, 15) and 3 different values for each of at least two other tree structure parameters (e.g., number of elements in leaf and tree depth). For all parameter combinations (there should be at least 27 combinations), perform a K-fold cross-

validation (k is freely chosen and can take values from 5 to 10). Provide plots for each set of parameters and compare the results. What set of parameters should be used in a random forest model? How does the model performance depend on the number of trees? Justify your answer.

c) Implement a random forest model using scikit-learn (RandomForestRegressor) or other library functions. Perform cross-validation with the parameters used in part b. Present box plots of the results obtained. Compare the results of parts b and c.

d) Prepare a model for predicting the target value for the "unseen" data:
   i) Construct a random forest model with the number of trees selected during the study and other tree parameters.
   ii) Apply the model to predict the target value of the test dataset (in original format). Calculate model accuracy metrics for the test dataset. Display at least 10 elements from the test dataset with predicted and actual values. **It must be possible to input the object values (in their original format) and get the value predicted by the model without training (e.g. in a separate file).**

1 Table. Information about datasets.

Dataset	Link
Medical Cost Personal Datasets	https://www.kaggle.com/datasets/mirichoi0218/insurance
Used Car Price Prediction	https://www.kaggle.com/datasets/therohithanand/used-car-price-prediction
Melbourne Housing Market	https://www.kaggle.com/datasets/anthonypino/melbourne-housing-market
Philippines Housing Market (Housing_v2.csv)	https://www.kaggle.com/datasets/klekzee/phillipines-housing-market?select=Housing_v2.csv
Quebec Housing Sales price	https://www.kaggle.com/datasets/rickyjackson/quebec-housing-sales-price

2 Table. Individual tasks.

No.	Internal code	Dataset*	Target variable
1	F0433	Medical Cost Personal Datasets	charges
2	F4857	Used Car Price Prediction	price_usd
3	F8623	Melbourne Housing Market	Price
4	F8633	Philippines Housing Market (Housing_v2.csv)	Price
5	F0494	Quebec Housing Sales price	sale_price
6	E8117	Medical Cost Personal Datasets	charges
7	F0221	Used Car Price Prediction	price_usd
8	F8794	Melbourne Housing Market	Price
9	F8810	Philippines Housing Market (Housing_v2.csv)	Price
10	F4640	Quebec Housing Sales price	sale_price
11	F8679	Medical Cost Personal Datasets	charges
12	F8683	Used Car Price Prediction	price_usd
13	F0479	Melbourne Housing Market	Price
14	E8979	Philippines Housing Market (Housing_v2.csv)	Price
15	F0104	Quebec Housing Sales price	sale_price
16	F8779	Medical Cost Personal Datasets	charges
17	F4639	Used Car Price Prediction	price_usd
18	F0686	Melbourne Housing Market	Price
19	F8723	Philippines Housing Market (Housing_v2.csv)	Price
20	F0438	Quebec Housing Sales price	sale_price
21	F8796	Medical Cost Personal Datasets	charges
22	F8761	Used Car Price Prediction	price_usd
23	F4890	Melbourne Housing Market	Price
24	F0133	Philippines Housing Market (Housing_v2.csv)	Price
25	E8948	Quebec Housing Sales price	sale_price
26	E9194	Medical Cost Personal Datasets	charges
27	E7961	Used Car Price Prediction	price_usd
28	F0484	Melbourne Housing Market	Price
29	F7320	Philippines Housing Market (Housing_v2.csv)	Price
30	F8785	Quebec Housing Sales price	sale_price
31		Medical Cost Personal Datasets	charges
32		Used Car Price Prediction	price_usd

* please check if the dataset provided here is the same as given in the first laboratory work (if not, use the same dataset as in the first task).